

Why a Simple Threshold Rule Is Hard to Beat: Volatility Regime Prediction Under Information Delay

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Independent

16 September 2026

Abstract

We ask how quickly the ability to predict a volatility regime decays when the forecaster's information is δ days out of date. On nine years of daily S&P 500 data we evaluate four forecasters against a binary target, namely whether realised variance five days hence exceeds its trailing 252-day median, at delays of zero to 55 trading days. A trivial rule, "is volatility above its median now", stays clearly above a constant forecast out to roughly one week of delay and is indistinguishable from one beyond two weeks. Neither the Yang-Zhang estimator (+2.0 points at zero delay, 95% CI $[-3.3, +7.3]$) nor a walk-forward HAR classifier (-4.0 points, $[-11.7, +3.2]$) improves on it at any delay. We explain that instead of reporting it: for a median-split target, under approximate joint lognormality, the threshold rule is the Bayes rule among functions of the delayed state alone. Measured accuracies sit at 90–106% of that restricted-information Gaussian benchmark, exceeding it at eight of ten delays, so Section 5 treats it as an approximation, never a ceiling. The headroom a better model could occupy is small relative to sampling uncertainty, so the negative result is about the target, not the models. Three robustness checks follow, on the loss, the windows and the split. The third breaks the median-split theory: at an upper quantile the Bayes cut becomes z_q/ρ instead of zero, and at the 90th percentile the Bayes rule itself attains only 90.1% against a majority-class 90.0%.

1. Introduction

A forecaster whose data arrives late is in a different position from one whose data is merely noisy. The question of how much predictive content survives a delay of δ periods is distinct from, though in population equivalent to, the familiar question of multi-step forecasting, and it has an obvious practical reading: how much is timely data worth?

This note reports a small, fully reproducible experiment on that question, and a negative result. The contribution is not a method. It is a measurement, a set of baselines that the measurement is compared against, and an argument for why the measurement could not have come out any other way.

Two things motivated the exercise. First, the volatility-forecasting literature almost never reports what was attainable, only what was attained; Andersen and Bollerslev [1] are the exception, and their point, that a low R^2 may be a property of the target and no failure of the model, is the point made here in a different setting. Second, claims about model sophistication under delay are usually made without a trivial baseline in view, and trivial baselines on persistent series are unusually strong.

2. Data

Daily open, high, low and close for the S&P 500 index, 12 September 2017 to 11 September 2026, 2,262 trading days, from a retail export. The series was reordered to ascending and validated: dates strictly increasing, high not below low, high not below either open or close, low not above either, prices

strictly positive. No survivorship or revision adjustment is possible with this source, which is a limitation recorded in Section 6.

3. Method

3.1 Volatility proxies

Two proxies are computed over rolling five-day windows. The close-to-close proxy is the mean of squared log returns over the window. For the second, write O_s , H_s , L_s and C_s for the open, high, low and close of session s , and $o_s = \log(O_s / C_{s-1})$, $c_s = \log(C_s / O_s)$, $u_s = \log(H_s / O_s)$, $d_s = \log(L_s / O_s)$ for the overnight, open-to-close, high and low log returns. The Yang-Zhang proxy combines overnight, open-to-close and Rogers-Satchell [7] components,

$$\sigma_{YZ}^2 = \sigma_o^2 + k \sigma_c^2 + (1 - k) \sigma_{RS}^2, \quad k = 0.34 / (1.34 + (n+1)/(n-1)), \quad (1)$$

where σ_o^2 and σ_c^2 are the sample variances of o_s and c_s over the n sessions ending at t , σ_{RS}^2 is the mean over them of $u_s(u_s - c_s) + d_s(d_s - c_s)$, and k is the weight for which the sum is minimum-variance and drift-independent under a Brownian price, with $n = 5$, following Yang and Zhang [8]. The close-to-close proxy discards the overnight gap entirely; the comparison between the two is therefore a direct test of whether that gap carries regime information.

3.2 Target

Let v_t be the close-to-close proxy and M_t its median over the trailing 252 days, computed using data up to t only. The target is

$$y_t = \mathbf{1}\{v_{t+h} > M_t\}, \quad h = 5. \quad (2)$$

A forecaster who is δ days late knows the proxy only to $t - \delta$, and therefore knows $M_{t-\delta}$, not M_t . Comparing a delayed observation against a contemporaneous threshold is a look-ahead error; we avoid it, and measure its size in Section 4.3.

3.3 Forecasters

Four forecasters, each seeing only information dated $t - \delta$ or earlier. *Majority* predicts the class that was more frequent in training. *Persist* predicts $\mathbf{1}\{v_{t-\delta} > M_{t-\delta}\}$, in close-to-close and Yang-Zhang variants. *HAR* is a logistic regression on the daily, weekly and monthly averages of the log proxy, the heterogeneous-autoregressive structure of Corsi [3] cast as a classifier, fitted by iteratively reweighted least squares on a rolling 1,250-day window, refitted every 21 days. Its training pairs are restricted to labels already resolved at $t - \delta$, so no label information crosses the information boundary.

3.4 Evaluation

Walk-forward over 694 test days, 27 November 2023 to 2 September 2026, after the 252-day median warm-up, the 1,250-day training window and the 55-day maximum delay are set aside. Because the five-day targets overlap, ordinary binomial intervals would be too narrow; we use a moving-block bootstrap [6] on the correctness series with block length $2h = 10$ and 2,000 resamples, and the same procedure on paired differences.

4. Results

Table 1 reports out-of-sample accuracy. The base rate in the test window is 0.468, so a constant "no" forecast would have scored 53.2%, but that constant is chosen with hindsight. A forecaster at the start of the window does not know which way the balance will fall, and choosing from the training window gives two answers depending on its shape: a rolling 1,250-day window picks "no" at every refit and also scores 53.2%, while the expanding window behind the majority rule in Table 1 picks the other class and scores 46.8%. Both are honest and they differ by 6.3 points, which is wider than several of the model differences discussed below. We take 53.2% as the hurdle because two of the three constructions agree on it, and report the expanding-window rule in Table 1 so that a reader can see the other. The walk-forward majority rule scores 46.8% because the training-window majority was the opposite class, which is honest walk-forward behaviour and no error.

Table 1. Out-of-sample accuracy (%) by delay, with 95% moving-block bootstrap intervals.

δ	majority	persist-CC	persist-YZ	har-CC	har-YZ
0	46.83	65.85 [58.6, 71.9]	67.87 [61.1, 73.3]	61.82 [53.9, 68.4]	68.59 [62.0, 74.5]
1	46.83	65.56 [58.5, 71.2]	66.71 [59.9, 73.2]	61.53 [53.5, 68.4]	67.44 [60.1, 73.5]
2	46.83	64.41 [57.6, 70.0]	65.56 [58.5, 72.0]	61.24 [53.2, 68.2]	66.57 [59.5, 72.2]
3	46.83	62.10 [55.0, 68.2]	63.26 [55.8, 69.5]	60.81 [53.2, 67.7]	64.84 [58.1, 71.0]
5	46.83	60.23 [53.3, 66.4]	60.95 [53.6, 67.3]	58.50 [50.4, 65.3]	60.95 [53.9, 67.6]
8	46.83	56.34 [49.4, 63.3]	57.20 [49.6, 63.4]	54.76 [47.3, 62.0]	57.20 [49.4, 63.8]
13	46.83	52.88 [46.1, 59.5]	54.90 [47.7, 61.0]	52.74 [45.1, 59.9]	53.75 [46.1, 60.5]
21	46.83	49.86 [43.7, 56.9]	50.00 [42.9, 57.4]	54.03 [46.4, 62.1]	51.87 [44.2, 59.2]
34	46.83	50.43 [43.8, 56.5]	49.86 [43.1, 56.5]	50.43 [42.2, 56.8]	47.84 [40.2, 54.6]
55	46.83	48.27 [40.8, 55.0]	44.24 [37.5, 51.3]	46.54 [38.8, 54.0]	46.25 [38.6, 53.3]

Every column falls with delay and they fall at different rates, which is the whole content of the table. The two persistence rules and the HAR classifier all start near 66–69% at zero delay and converge on the majority rule's 46.83% by three weeks; by $\delta = 21$ no forecaster's interval excludes the 53.2% constant-forecast hurdle. The intervals are wide, roughly seven accuracy points either side, so what the table establishes is the decay, not an ordering among the four. The comparisons that need an ordering are made on paired differences in Table 2, where the common days cancel and the intervals narrow.

4.1 The decay

Accuracy falls from about 68% at zero delay to the constant-forecast level by roughly two weeks. Taking 53.2% as the reference, the threshold rule is clearly above it through $\delta = 5$, marginal at $\delta = 8$, and indistinguishable from it at $\delta = 13$ and beyond. The apparent non-monotonicity between $\delta = 13$ and $\delta = 21$ is within noise: those two accuracies differ by less than one point against a bootstrap standard error of roughly five on the difference.

4.2 Neither refinement helps

Table 2. Paired accuracy differences (percentage points) with 95% block-bootstrap intervals. No interval excludes zero.

δ	har-CC – persist-CC	har-YZ – persist-YZ	persist-YZ – persist-CC
0	−4.03 [−11.7, +3.2]	+0.72 [−4.9, +6.9]	+2.02 [−3.3, +7.3]

δ	har-CC – persist-CC	har-YZ – persist-YZ	persist-YZ – persist-CC
3	−1.30 [−8.2, +5.2]	+1.59 [−3.6, +7.2]	+1.15 [−4.0, +6.3]
5	−1.73 [−8.6, +4.6]	+0.00 [−5.3, +5.3]	+0.72 [−4.3, +5.5]
8	−1.59 [−8.5, +4.8]	+0.00 [−5.2, +5.0]	+0.86 [−4.8, +5.6]
13	−0.14 [−6.6, +6.5]	−1.15 [−6.9, +4.5]	+2.02 [−3.2, +6.3]
21	+4.18 [−3.2, +11.1]	+1.87 [−4.2, +7.8]	+0.14 [−5.5, +5.0]

The Yang-Zhang proxy is worth about two accuracy points at zero delay and less thereafter, never distinguishable from zero. The HAR classifier is worse than the threshold rule at short delays and no better at long ones. On this target, over this sample, the overnight gap and the multi-horizon regression structure both fail to earn their place.

4.3 The look-ahead leak is negligible

Comparing the delayed proxy against M_t rather than $M_{t-\delta}$, the error described in Section 3.2, changes accuracy by at most 0.43 points at any delay where the rule has skill, and at $\delta = 34$ it lowers accuracy by 1.73. The leak is real and should be avoided, but it explains nothing.

5. Why the threshold rule is a strong benchmark

A negative result is only interesting with a mechanism attached. Here there is one, and it is close to trivial once stated.

Write the two decision variables in log form relative to their own thresholds: $a = \log(v_{t-\delta} / M_{t-\delta})$ for the forecaster, $b = \log(v_{t+h} / M_t)$ for the outcome. The target is $\mathbf{1}\{b > 0\}$ and the threshold rule is $\mathbf{1}\{a > 0\}$. If (a, b) are approximately jointly normal with correlation ρ and near-zero medians, then $P(b > 0 | a)$ is monotone in a and crosses one half at $a = 0$, and the threshold rule is the Bayes rule *among functions of a alone*. The accuracy it then attains is

$$A^*(\delta) = 1/2 + \arcsin \rho(\delta) / \pi. \quad (3)$$

Two things must be said about that object before Table 3 is read, because an earlier draft said neither and called it a ceiling.

It is not a ceiling for the models compared against it. The HAR classifier sees three features, the current delayed state and its five- and twenty-two-day means, and making the sign rule optimal over that larger set needs a sufficiency condition, $P(b > 0 | a, w, m) = P(b > 0 | a)$, which bivariate normality of (a, b) does not supply. We tested it: a logistic on all three features beats a logistic on a alone at none of the ten delays at $|z| > 1.96$, so the condition is not rejected here, and the extra lags are not where the excess over A^* comes from.

What does fail is the centring assumption, and our own data rejects it. Estimating $P(b > 0 | a)$ non-parametrically on deciles of a gives a function that is not monotone and that crosses one half at $a = +0.27$ instead of at zero. The rule's threshold is in the wrong place. That is why a logistic free to choose its own threshold beats the sign rule at several short delays, and why measured accuracy exceeds A^* at eight of the ten delays below. A quantity the process can exceed is an approximation and not a bound, so we call A^* a *restricted-information Gaussian benchmark* and claim nothing stronger for it.

Table 3 evaluates it on the same test window.

Table 3. Attainable accuracy under the normal approximation, against measured accuracy.

δ	$\rho(a, b)$	Gaussian benchmark A^*	persist-YZ	har-YZ	% of benchmark
0	0.4502	64.87	67.87	68.59	104.6
1	0.4039	63.23	66.71	67.44	105.5
3	0.3179	60.30	63.26	64.84	104.9
5	0.2539	58.17	60.95	60.95	104.8
8	0.1879	56.02	57.20	57.20	102.1
13	0.1712	55.48	54.90	53.75	99.0
21	0.1741	55.57	50.00	51.87	90.0
55	-0.1202	46.17	44.24	46.25	95.8

The measured accuracies sit between 90% and 106% of the benchmark. The excess above 100% at short delays is expected and instructive: log realised variance has heavier tails than the normal approximation allows, which makes the real problem slightly easier than the bound, and the 694-day sample gives intervals of roughly seven points either way. The reading is not that the rule exceeds

what is possible, but that it is at the boundary, and that the normal approximation is the binding assumption, not the models.

This is the substance of the negative result. Estimated headroom is small relative to sampling uncertainty, and neither refinement we evaluated exploits it reliably, so their failure to improve on the threshold rule is not evidence that those tools are weak. It is a property of the target: a median-split label discards exactly the magnitude information that a better estimator or a richer regression would have supplied. A study wanting to discriminate between volatility models should not use one.

6. Robustness: three things the design fixed by fiat

Three choices in Sections 3 and 4 were made once and never varied: accuracy as the loss, 252 days as the median window with 1,250 as the training window, and the median itself as the split. Each was reasonable and none was tested. This section tests all three. Two of the answers support the note; the third says the question cannot be settled on nine years of data, and says so instead of reporting a number.

6.1 Accuracy is a coarse loss, so use sharper ones

Accuracy sees only which side of a half a forecast fell on. Two sharper measures are available. AUC asks whether a forecaster orders the days correctly and is invariant to any monotone transform, so it can score a rule that has a margin behind it even when that rule emits no probability: the persistence rules have one, the decision variable $a = \log(v_{t-\delta} / M_{t-\delta})$ whose sign is the rule. The Brier and log scores measure the quality of a probability, are strictly proper, and cannot be computed for a forecaster that emits a hard label at all.

Table 4. The same forecasters under three measures. AUC(margin) ranks by the continuous decision variable a forecaster holds; AUC(label) ranks by the 0/1 it emits, and for a binary forecast equals balanced accuracy. The base-rate row is the recursive base rate, the probability a forecaster could actually have quoted; its AUC is below one half because a nearly constant series is ranked by its own slow drift, and it should not be read as a chance benchmark, which is 0.500.

δ	forecaster	accuracy	AUC (margin)	AUC (label)	Brier	log
0	persist-YZ	67.87%	0.735	0.681	n/a	n/a
0	har-YZ	68.59%	0.705	0.674	0.2195	0.6357
5	persist-YZ	60.95%	0.660	0.611	n/a	n/a
5	har-YZ	60.95%	0.628	0.598	0.2470	0.6997
21	persist-YZ	50.00%	0.535	0.501	n/a	n/a
21	har-YZ	51.87%	0.508	0.507	0.2849	0.7890
21	recursive base rate	42.94%	0.336	0.409	0.2513	0.6958

The objection is answered in the opposite direction to the one it anticipated. Under accuracy the two forecasters are inseparable, as Section 4.2 reported. Under AUC they separate at seven of ten delays, and in the threshold rule's favour every time, never the HAR classifier's. At $\delta = 0$ the gap is -0.0299 with a block-bootstrap interval of $[-0.0542, -0.0066]$; the pattern holds out to $\delta = 13$ and fades after. The raw margin orders the days better than the fitted logistic's probability does, so the claim of parity in Section 4.2 is, if anything, too generous to the refinement.

Two further readings are worth stating because accuracy cannot reach either. The first is what discretising costs: persist-YZ's margin out-ranks its own emitted label at nine of ten delays, by as much as 0.055 of AUC at $\delta = 0$. A forecaster who reports the sign throws that away, and Table 1 cannot see it. The second is less comfortable for the note. Beyond about three weeks the HAR classifier's *probability* is worse than the recursive base rate on Brier score, significantly so at $\delta = 21, 34$ and 55, while its accuracy at those delays still looks respectable. A user who needs a probability instead of a label should read the Brier column, and the Brier column says the model has stopped being useful earlier than the accuracy column suggests.

6.2 The two windows

The median window is not a free parameter of the forecaster; it defines the target, since $y_t = 1$ exactly when the proxy five days on exceeds M_t . Changing it changes the question. A 63-day median asks whether volatility will exceed its recent quarter's typical level and a 504-day median asks about a two-year level, and those have different base rates and different intrinsic difficulty. Accuracies are therefore not comparable across windows. What is comparable is the contest within each window, and that is what the note's claim is about. Every cell is scored on one common 476-day window, whatever the most demanding cell can afford, so no difference between cells is a difference in sample.

Table 5. Accuracy of the HAR classifier minus the threshold rule, in percentage points, across the window grid, with 95% block-bootstrap intervals. Positive would favour the HAR classifier. Every cell on the same 476 test days.

median	training	$\delta = 0$	$\delta = 3$	$\delta = 5$	$\delta = 13$	$\delta = 21$
63	1250	-5.25	-5.88	-7.14	-0.63	+3.36
126	1250	-7.35	-6.93	-8.19	-2.94	-2.31
252	500	-4.41	-6.09	-5.46	-6.30	-2.94
252	1250	-1.89	-2.52	-2.94	-5.46	-1.47
504	1250	+0.63	-0.84	-2.31	-2.10	-2.31

Across the full grid of four median windows and two training windows at five delays, forty cells in all, the HAR classifier significantly beats the threshold rule in none, and the threshold rule significantly beats the HAR classifier in fifteen. The note's negative result is not a property of 252 days: it survives a fourfold change in the median window and a 2.5-fold change in the training window, and the direction never reverses.

One reading of the same grid does not survive, and we report it because an earlier draft of this section would have got it wrong. The delay at which the rule stops beating a constant is later for short median windows, five days at 63 and 126 against zero days at 252 and 504, which looks like a finding about window length. It is partly a loss of power. Re-running the note's own cell on the largest window it can afford, 728 days instead of the common 476, moves that boundary back out from zero to three days. Shorter median windows also give a target closer to a 50/50 split, which is easier to beat by construction. The ranking across windows stands; the absolute boundary in Section 4.1 is specific to the note's own sample and should not be re-read off this table.

6.3 Off the median: a tail target

Volatility regimes are asymmetric, and a median split does not ask about spikes. The natural extension is to move the target to an upper quantile, and doing so breaks both halves of Section 5 in ways worth being explicit about.

The rule breaks first. For approximately joint normal a and b with correlation ρ , the Bayes rule against the target $1\{b > z_q\}$ predicts one when $P(b > z_q | a)$ exceeds a half, which happens exactly at

$$c^*(q) = z_q / \rho, \quad z_q = \Phi^{-1}(q) \quad (4)$$

At $q = 0.5$ this is zero and the note's rule is the Bayes rule, which is Section 5's claim. Away from the median it is not: at $q = 0.9$, with $\rho = 0.52$ at zero delay, the optimal cut is 2.46 standard deviations. Carrying "is volatility above its median now" to a tail target therefore tests a misspecified rule, and it would lose for a reason that has nothing to do with the question. The benchmark breaks second: with the two thresholds no longer both at zero there is no arcsin form, and the benchmark has to be integrated numerically. That integrator reproduces $\frac{1}{2} + \arcsin(\rho)/\pi$ at $q = 0.5$ to a largest absolute difference of $1.41\text{e-}07$, which is the check that licenses using it elsewhere.

Table 6. The Gaussian benchmark off the median, at $\delta = 0$ with $\rho = 0.521$. "Majority" is the accuracy of never predicting the event. The benchmark is the accuracy of the Bayes rule; the last column is the accuracy the note's median rule would attain against the same target.

q	z_q	$c^*(q)$	majority	benchmark	note's rule
0.50	0.000	0.00	50.0%	67.45%	67.43%
0.65	0.385	0.74	65.0%	71.06%	65.37%
0.75	0.674	1.29	75.0%	77.13%	62.42%
0.90	1.282	2.46	90.0%	90.11%	55.33%

The bottom row is the result, and it is a theoretical one that no amount of data would change. At the 90th percentile the Bayes rule itself attains 90.11% against a majority-class 90.00%. Even a forecaster who knows ρ exactly and applies the optimal cut beats "never predict a spike" by a ninth of a percentage point. Accuracy at a tail target is not a demanding measure that the threshold rule might fail; it is a measure on which no rule can demonstrate anything, and a paper reporting a 90% accuracy against a spike target would be reporting the base rate.

Measured on balanced accuracy instead, the mean of sensitivity and specificity, with chance at 0.500 whatever the base rate, the two rules change places depending on the objective. At fifteen of twenty quantile-delay cells the rule that maximises raw accuracy is not the rule that maximises balanced accuracy: the fitted cut is right more often by almost never calling a spike, while the note's median rule catches spikes at the cost of false alarms. Neither is better; they answer different questions, which is the referee's point about loss functions arriving in a different place from where it was aimed.

What we cannot do is say which rule predicts spikes best. On 728 test days the 90th-percentile target has 104 positive days and, with five-day overlap, on the order of ten independent episodes. Of fifteen pairwise comparisons at that quantile, the balanced-accuracy intervals separate one. The tail target is predictable: at $q = 0.9$ and $\delta = 0$ the median rule's balanced accuracy is 0.678 with an interval of $[0.608, 0.749]$, clear of chance. But this sample cannot rank the forecasters against it. Nine years does not contain enough independent volatility spikes, and no change of estimator fixes a shortage of events.

7. Limitations

The test window is 694 days and the intervals are wide; nothing here separates differences smaller than about five accuracy points. A single index over a single nine-year window cannot speak to other

markets or other regimes. The evaluation period is narrower still: it begins in November 2023 and contains no major macro transition, so the objection that a rolling median fails to adjust across a structural regime change cannot be tested here at all. That is a shortage of sample rather than of method. The companion paper, whose test window runs from May 2007 and contains 2008, 2020 and 2022, does test it, and finds the cross-sectional result strengthens in stress instead of breaking down. The price series is a retail export with no point-in-time guarantee, so any revision to the underlying index is silently inherited. The Gaussian benchmark of Section 5 is a normal approximation, and Table 3 shows it is not exact; Section 6.3 gives its generalisation away from the median, which has no closed form and is integrated numerically. The delay axis is applied to an information set assumed empty after $t - \delta$; a real late forecaster would still observe contemporaneous prices, so these numbers are a lower bound on what such a forecaster could do. Accuracy on a binary target is a coarse loss. Section 6.1 answers that within this design, using AUC and the Brier and log scores; for continuous volatility forecasting, QLIKE and the robust family of Patton [5] remain the appropriate measures, and the companion paper uses them. Finally, Section 6.3 leaves a question open instead of answering it: this sample contains too few independent volatility spikes to rank forecasters against a tail target, and that is a shortage of events and not of method. What would repair that is worth naming, because the obvious remedy is the wrong one. Sampling the same nine years more finely measures the same five-day target more precisely; it does not create independent tail episodes, whose number is set by the calendar. Redefining the target at an intraday horizon would create them, but it would be a different question. The episode count grows in only two honest ways: a longer panel, decades and not years, which for range-based estimators means data most researchers cannot obtain; or a wider one, since spikes accumulate across markets as well as through time, so a cross-section reaches a given episode count in a fraction of the calendar, and the companion paper's eight markets over nineteen years is the shape of design this question needs. A third route avoids the shortage without solving it: score forecasters against a target that PRICES the tail instead of waiting for it, since an option surface quotes a tail every day and a daily score needs no spike to arrive. Each of the three is a different paper, and none of them is this one.

8. Reproducibility

Every number in this note was produced by six scripts: `lab02_delay_curve.py`, `lab02b_threshold_ceiling.py`, `lab20_ceiling_is_not_a_ceiling.py`, `lab24_proper_scores.py`, `lab25_window_sensitivity.py` and `lab26_quantile_target.py`, all run on the data described in Section 2 with a fixed seed. All ship in the repository named in the main paper, are run by `run_all.py --check`, and are diffed line by line against stored reference output. The second imports the first instead of restating it, so the accuracy column of Table 3 and the accuracy columns of Table 1 cannot drift apart. A separate script, `verify_paper.py`, checks every row of Table 3 in this note against that reference output, and fails if any cell disagrees.

That check exists because of what it found. Until late in the writing, Table 3 came from a scratch script that read a price file from the author's desktop and compared the benchmark against accuracy figures pasted in as literals from an earlier run. Nobody else could have run it, and had `lab02` changed, the pasted column would have gone on agreeing with a number that no longer existed. Every figure in the table survived being recomputed properly, which is the good outcome and was not the guaranteed one.

Disclosure of AI use

Claude (Anthropic) was used throughout this work: to write the replication and execution scripts, to draft and revise the text, and to assist in the iterative design of several tests. All research questions, design decisions, and claims were directed and accepted by the author, who is solely responsible for the contents. Every number reported is produced by the scripts in the public repository and checked against them by `verify_paper.py`.

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